

Reinforcement Learning versus a Genetic Algorithm for Dynamic Ride Surge Pricing

Alabi Taiwo Oluwatamilore · x24335592

H9AIDM · MSc Artificial Intelligence · National College of Ireland

The problem as a Markov Decision Process

STATE S

24 states = 4 time x 2 weather x 3 demand

morning_rush 07-09 / day 10-15 /

evening_rush 16-19 / night

dry / wet -> low / mid / high

ACTION A

5 multipliers = { 1.0, 1.25, 1.5, 1.75, 2.0 }

TRANSITION P

weather and demand sampled from the data

one episode = one 24-hour day

action-independent: price does not change

the next state

REWARD R

**m x base_fare(s)
x rides_per_hour(s)
x acceptance(m, s)**

$\text{acceptance}(m, s) = \max(0, 1 - K(s)(m-1))$

base_fare = price / historical surge_multiplier

strips Uber's own surge out, so the agent
prices from a clean fare

$K(s)$ is the price sensitivity. It is assumed,
not learned. Smaller K = riders are more inclined to accepting
rides, bigger K = rides are more likely to ditch rides for
alternatives.

All 24 states are populated.

Thinnest holds 2,453 rides.

Two approaches to the same policy

Q-LEARNING (reinforcement learning)

learns by doing, one state at a time

Q-table = 24 x 5 = 120 values

$Q(s,a) \leftarrow Q(s,a) + a[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$

$a = 0.1$ apply 10% of the gap per update

$\gamma = 0.9$ near future weighted more

ϵ : 1.0 to 0.05, decay 0.999 per episode

3,000 episodes x 24 hours = 72,000 updates

Learns from a single transition, so it could keep adapting on a live platform.

GENETIC ALGORITHM (metaheuristic)

evolves whole policies at once

genome = 24 slots = one full policy

search space = 5^{24} = about 6×10^{16}

population 40, 60 generations

tournament $k=3$, uniform crossover

mutation 0.1 per slot, elitism 2

fitness = 10 days on fixed seed 7

Never scores a single decision, only whole policies. Needs a replayable simulator.

Baselines and results

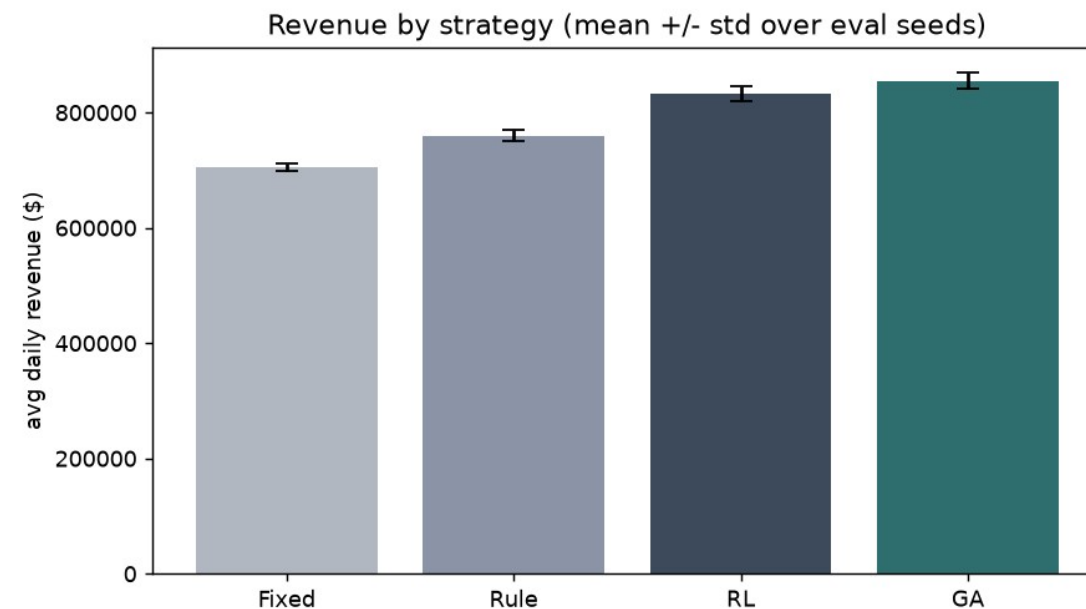
8 seeds x 30 days. Every strategy faces the identical seeded worlds.

Fixed baseline: always 1.0x, total revenue = 704,646

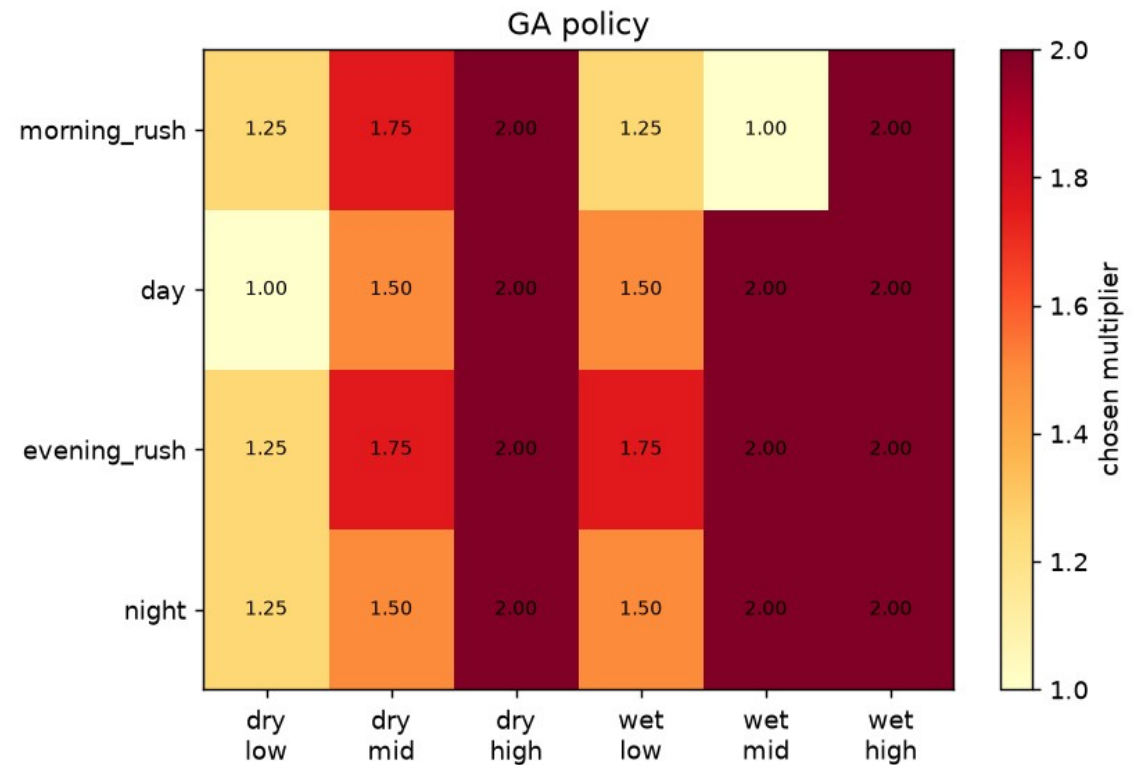
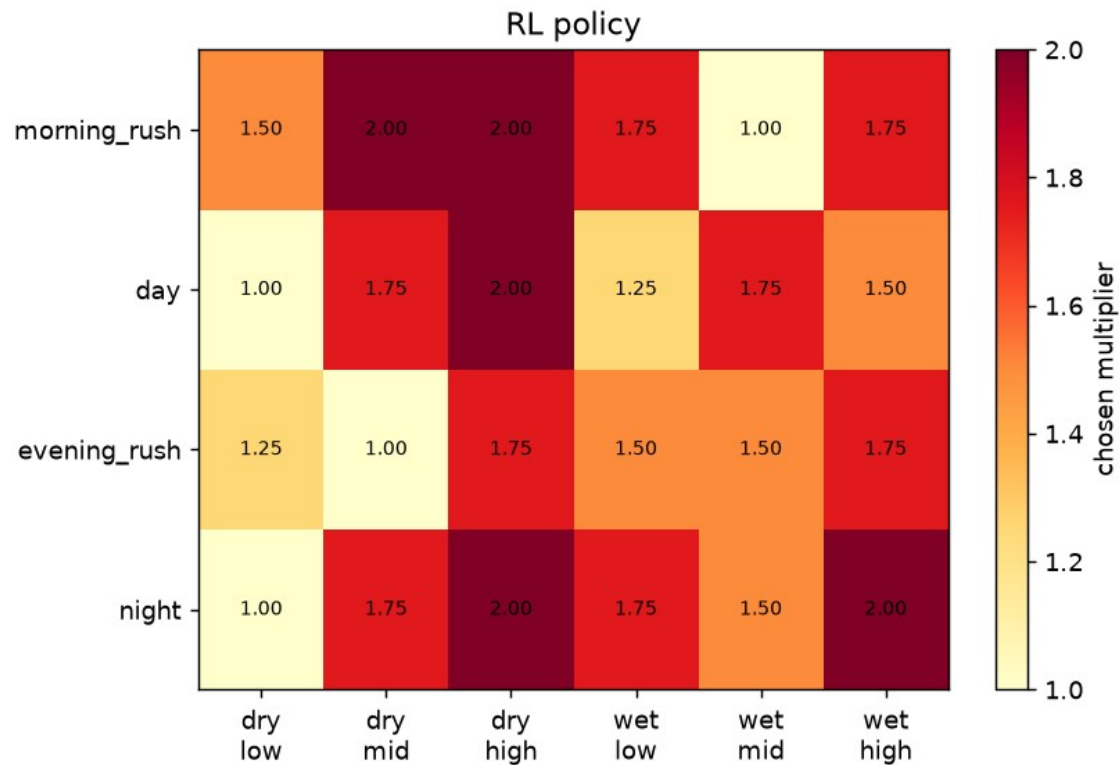
Rule baseline: 1.5x if wet or rush, total revenue = 760,117 +7.9% from fixed baseline

RL:Q-learning, total revenue = 833,103 +18.2%

GA: evolved, total revenue = 855,244 +21.4%



What both agents discovered on their own



Two different mechanisms, the same economic structure. Neither agent was given any pricing rule.

Limitations and future work

LIMITATIONS

K, the price elasticity, is assumed and not learned;

Its important because, k affects acceptance rate.

Also, the drivers are not modelled.

FUTURE WORK

Learn K instead of assuming it.

Model the supply side so surge rations demand and summons drivers.